

Executive Summary

Cell towers carry no condition sensors, so operators cannot identify fragile sites before one fails. This project derives tower condition indicators from open satellite, elevation, and infrastructure data, combines them into an interpretable Tower Health Risk Index with per-factor attribution, and converts that attribution into a dispatched maintenance schedule. The result is a shift from calendar-based and complaint-driven maintenance to exposure-based, pre-emptive intervention, using open global datasets that require no new data collection.

Build status. The detect → decide → dispatch chain is implemented end to end and runs on a real feature table for a Malaysian pilot area of interest (Sunway, Selangor; 132 operator-node sites). The indicators currently reaching the index are derived from elevation, hydrology, and open infrastructure data. The imagery-segmentation perception stage described in §4.2 is designed and interface-wired but not yet executed; §3 marks precisely which sources are live and which are readiness-assessed only. Nothing in this document claims a component that is not in the repository.

1.0 Problem Statements

Cell towers are critical infrastructure with no condition sensors. Operators cannot see which towers are fragile until one fails. Tower outages are driven by the environment around the site: flooding, unstable slopes, lightning, and grid dependence. Maintenance is nonetheless scheduled by calendar cycle or by complaint, because no operator holds a geospatial risk profile of its own sites. Four consequences follow:

1. **Reactive repair.** Crews are dispatched after the outage, when access roads may already be cut. [1],[2]
2. **Misallocated hardening.** Budgets follow asset age and past incidents rather than exposure. A three-year-old tower in a flood plain is invisible to a calendar-based programme. [3]
3. **Longest outages hit thinnest coverage.** Rural single-tower areas, where an outage isolates a community, are the slowest to reach. [4]
4. **A ranked list is not a decision.** A risk score alone does not state what to fix, how soon, or who to send. [5]

The gap: every driver that matters is observable from open satellite and elevation data, yet exists in no operator table, and nothing today carries it through to a dispatched crew.

1.1 Needs Statements

Need	Feature
See tower exposure before the season, without site sensors	Risk index over remotely-sensed drivers: flood, terrain, equipment, grid

Rank by exposure rather than age, and name the driving mechanism	Noisy-OR scoring with per-factor attribution
Surface the population actually affected, so planners can break ties between equally risky towers	Gridded population exposure per site, reported alongside risk and never inside it
Convert the reason a tower is risky into action, crew, parts, and date	Agentic Maintenance Scheduler

Scope of prediction: *The system predicts maintenance need, priority, and urgency: which towers warrant intervention, driven by which failure mechanism, and how soon.*

2.0 Target Users

	Who	Benefits
Primary User	Field-maintenance planners at ASEAN operators	What to inspect, why, and when, rather than a score.
Secondary User	Network-operations planners	Evidence for crew sizing and hardening-capex prioritisation
Direct consumer	Field crews	A work order carrying action, parts, crew type, and the reason
Tertiary user	Regulators (MCMC) and disaster-management agencies	Auditable infrastructure-resilience measure for service-obligation areas

3.0 Datasets and Data Sources

Layer	Source	Role
Tower sites and radio generation	Urban Multi-Operator QoE-Aware Dataset, Sunway (Mendeley 10.17632/dx5xyy fz2y.1)	Asset inventory and equipment-age proxy
Terrain	Copernicus DEM GLO-30 (Planetary Computer STAC)	Slope, ruggedness (TRI), height-above-drainage (HAND)
Surface water	Sentinel-2 L2A imagery with OpenStreetMap water	Flood-driver geometry, distance-to-water
Exposure, dense areas	Overture Maps buildings	Building counts and footprint area
Exposure, rural "dark" zones	Sentinel-2 imagery	Footprints extracted where map coverage is sparse

Lightning	NASA LIS/OTD flash-rate climatology	Strike density per tower
Grid dependence	VIIRS night-lights with OSM power infrastructure	Electrification and distance-to-grid
Land cover (context)	ESA WorldCover	Built-up versus vegetation and water setting

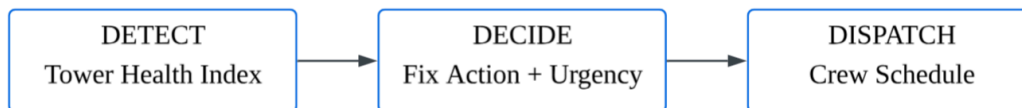
Data Contract: All perception outputs resolve to one row per tower: *coordinates*, *radio type*, *hand_m*, *dist_water_m*, *slope_deg*, *tri*, *flash_density*, *dist_power_m*, and *exposed_pop*, cached as GeoParquet or CSV. Nothing is fetched live during the demonstration.

Declared Assumptions: Crew counts, shift hours, travel speeds, and part lead times are demonstration parameters, declared as assumptions in the user interface and in this document.

4.0 Proposed GeoAI Approach

4.1 Core Idea: Replacement of Missing Sensors with Remote Sensing

A tower's environment is observable from space even when the tower itself is not instrumented: the water that floods it, the slope it stands on, the storms that strike it, and the grid it depends on. Deep learning converts that imagery and elevation data into condition indicators, and those indicators drive the maintenance decision. This substitution is the reason geospatial AI is load-bearing in this system rather than incidental.



4.2 Perception Layer

Pretrained inference via *opengeos/geoai*:

- **Water segmentation** (U-Net or SAM family) on Sentinel-2, giving flood exposure, combined with HAND derived from the DEM.
- **Building-footprint extraction** (Mask R-CNN family), giving the population affected by an outage in rural zones where no building database exists. Reported as consequence and kept out of the risk score (§4.3).
- **DEM derivatives**, giving slope and ruggedness as proxies for landslide risk, wind load, and access difficulty.
- **Fallback:** text-prompted GroundedSAM or CLIPSegmentation if a model is unavailable.

These outputs produce the risk drivers that exist in no operator table. Everything downstream is transparent arithmetic and optimisation.

4.3 Layer 1: Tower Health Risk Index (Detect)

1. **Membership functions:** Each raw factor maps to a failure contribution $p \in [0,1]$ through a monotonic curve with a physically justified shape. Flood risk, for example, rises steeply as height-above-drainage falls below approximately 5 m, then saturates. These are not linear min-max scalings.
2. **Noisy-OR combination:** $\text{Risk} = 1 - \prod(1 - p_i \cdot w_i)$, the probability that at least one driver triggers failure. It is bounded and monotonic, and any single severe factor dominates, which matches weakest-link failure behaviour. A weighted average would incorrectly average a new tower in a severe flood zone down to "safe".
3. **AHP weights** (Saaty pairwise, consistency ratio $\text{CR} < 0.1$) or published reliability literature. Weights are documented rather than hand-tuned.
4. **Per-factor attribution** (leave-one-out), for example: Tower MY_1042: 0.82 (flood 0.41, power 0.28, terrain 0.19, equipment 0.12).

Scores resolve into a maintain / watch / OK band with an associated urgency.

Likelihood and consequence are kept separate. *risk* answers how likely a tower is to fail, covering hazard and vulnerability only. *exposed_pop* answers how much it matters if it does. Folding population into the noisy-OR would conflate the two and render the score uninterpretable, so exposure is reported as its own column and drawn as its own map layer. Where the two must be combined for ordering, the combination is explicit and visible:

$$\text{priority} = \text{risk} \times \log1p(\text{exposed_pop})$$

The two questions, whether a tower is fragile and whether it matters, are therefore answered by two separate numbers.

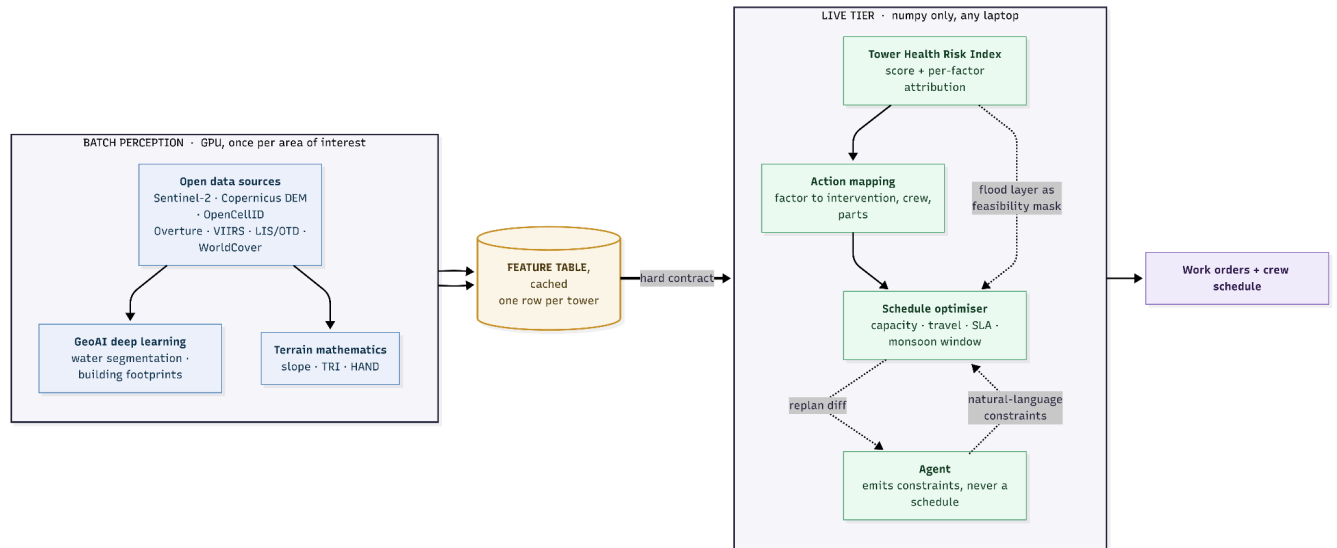
4.4 Layer 2: Agentic Maintenance Scheduler (Decide & Dispatch)

Attribution is the hinge. A tower that is risky because of flood needs drainage and a raised cabinet, requiring a civil crew and weeks of lead time. The same score driven by equipment generation needs a radio refresh and a capex line. Identical score, different crew, different parts, different timeline.

- **Action mapping.** A deterministic lookup from dominant factor to intervention, crew type, parts, and lead time. The *why* string is copied verbatim from attribution, so the explanation chain runs unbroken from raw DEM pixel to the sentence a crew reads.
- **Schedule optimiser.** Minimises $\Sigma \text{risk} \times \text{days-until-serviced} + \lambda \cdot \text{travel}$, subject to crew capacity, travel time, part availability, SLA caps, and a **monsoon weather window** that re-reads the index's own flood layer as a feasibility mask. This is the point at which the two layers interlock rather than merely chain.

- **Agent.** The LLM performs only the task no solver can: converting a planner's sentence, such as "the Kelantan road is washed out", into a constraint, and explaining what changed after a replan. **The model never emits a schedule; it emits constraints, and a solver emits the schedule.**

4.5 Architecture



Perception is precomputed, so a weight slider can be adjusted and both the ranking and the schedule update instantly, and no live model failure can interrupt the demonstration. The dotted edge from the index to the scheduler is the interlock described in Section 4.4: the same flood layer that raises a tower's risk also constrains when that tower can be serviced.

5.0 Expected Outcomes, Impact and Scalability

5.1 Deliverables

- Ranked tower-risk table with uncertainty band, per-factor attribution, and exposed population as a separate column.
- Interactive map showing risk and exposure, with a per-tower breakdown, recommended action, parts, and scheduled date.
- Work order per at-risk tower: action, crew type, parts, urgency, and the attribution sentence justifying it.
- Crew schedule (crew $\times$ day) with agent chat for natural-language constraints and replan diffs.
- Rank-stability figure and the policy-comparison metric.

5.2 Impact

- **Shifts maintenance from reactive to pre-emptive**, enabling inspection before the monsoon rather than after the outage.
- **Redirects capex by exposure rather than by age**, surfacing the new tower in a flood plain that calendar inspection misses entirely.

- **Closes the loop to action**, so risk becomes a dispatched crew with the correct parts rather than a report.
- **Reduces outage duration in rural areas**, where a single failure isolates a community.
- Provides **regulators with an auditable resilience measure** for service-obligation areas.

5.3 Regional Fit

The system is built on a Malaysian area of interest with Kelantan as the reference region. Its northeast-monsoon flooding (November to March) exercises the flood pathway that dominates the model, and the same seasonality re-enters the scheduler as a constraint, since civil work is not dispatched to flood-zone towers during monsoon.

5.4 Scalability

- **Zero new data collection.** Every input is open and global, and the pipeline reruns for any ASEAN Member State by changing a bounding box.
- **Cheap to refresh.** Perception runs once per area of interest on GPU; scoring and scheduling are numpy-only and re-run in seconds, so risk can be updated seasonally or after a flood event.
- **Transferable beyond telecom.** The hazard and vulnerability structure applies unchanged to substations, water pumps, clinics, and road assets.
- **Operator-extensible.** If an operator supplies even sparse real outage records, the same weights can be calibrated by weak supervision, an upgrade path that does not require rebuilding the system.

6.0 Limitations

- **Data:** Open sources are proxies, not asset records. Crowd-sourced tower positions, 30 m DEM, radio generation as an age proxy. Handled by sampling over an uncertainty buffer and never relying on a single factor.
- **Method:** No failure labels exist, so we claim robustness, not accuracy. Rank stability proves the ranking is not arbitrary; it does not prove the weights are right. Correctness needs operator failure data, the stated calibration path.
- **Scope:** Crew counts and lead times are demo assumptions, so the claim is the policy delta, not absolute timings.

7.0 Reference

- [1] S. K. Amir, "Malaysia's approach to digital justice: Internet access as a human right," *OpenGlobalRights*, Mar. 24, 2026. [Online]. Available: [OpenGlobalRights](#) [Accessed: August 12, 2026].
- [2] BERNAMA, "Sabah floods: MCMC restores 14 telecommunications towers," *BERNAMA*, Jul. 2, 2024. [Online]. Available: [BERNAMA](#) [Accessed: August 12, 2026].

[3] BERNAMA, "Fahmi: 11 Sabah west coast telecom towers to get urgent site hardening ahead of monsoon," *Malay Mail*, Sep. 26, 2025. [Online]. Available: [Malay Mail](#) [Accessed: August 12, 2026].

[4] T. Crljen, "How predictive maintenance drives cost savings," *WorkTrek*, Sep. 17, 2025. [Online]. Available: [WorkTrek](#) [Accessed: August 12, 2026].

[5] Innover Digital, "Truck roll optimization insights for a leading US-based telecom major," *Innover Digital*, n.d. [Online]. Available: [Innover Digital](#) [Accessed: August 12, 2026].